

Wet deposition of fission-product isotopes to North America from the Fukushima Dai-ichi incident, March 2011.



Using the infrastructure of the National Atmospheric Deposition Program (NADP), numerous measurements of radionuclide wet deposition over North America were made for 167 NADP sites before and after the Fukushima Dai-ichi Nuclear Power Station incident of March 12, 2011. For the period from March 8 through April 5, 2011, wet-only precipitation samples were collected by NADP and analyzed for fission-product isotopes within whole-water and filterable solid samples by the United States Geological Survey using gamma spectrometry. Variable amounts of $(^{131})\text{I}$, $(^{134})\text{Cs}$, or $(^{137})\text{Cs}$ were measured at approximately 21% of sampled NADP sites distributed widely across the contiguous United States and Alaska. Calculated 1- to 2-week individual radionuclide deposition fluxes ranged from 0.47 to 5100 Becquerels per square meter during the sampling period. Wet deposition activity was small compared to measured activity already present in U.S. soil. NADP networks responded to this complex disaster, and provided scientifically valid measurements that are comparable and complementary to other networks in North America and Europe.